PROGRAM:

```
import java.util.Scanner;
public class ElectBill
{
    public static void main (String[] args)
    {
        Consumer ob = new Consumer();
        ob.Getdata();
        ob.Calc();
        ob.Display();
    }
}
class Consumer
{
    Scanner in = new Scanner (System.in);
    Scanner ins = new Scanner (System.in);
    int cno;
    String cname,type_of_conn;
    double pre_reading,curr_reading,unit_consumed,tbill;
    void Getdata()
    {
        System.out.print ("\n\t Enter Consumer number = ");
        cno = in.nextInt();
        System.out.print ("\n\t Enter Consumer name = ");
        cname = ins.nextLine();
        System.out.print ("\n\t Enter the type of connection = ");
        type_of_conn = ins.nextLine();
        System.out.print ("\n\t Enter pre_month reading = ");
        pre_reading = in.nextDouble();
        System.out.print ("\n\t Enter current_month reading = ");
        curr_reading = in.nextDouble();

    }
    void Calc()
    {
        unit_consumed = curr_reading - pre_reading;
        if(type_of_conn.contains("domestic"))
        {
            if (unit_consumed <= 100)
                tbill=1* unit_consumed;
            else if (unit_consumed >100 && unit_consumed <= 200)
                tbill=2.50* unit_consumed;
            else if (unit_consumed >200 && unit_consumed <= 500)
                tbill=4* unit_consumed;
            else
                tbill=6* unit_consumed;
```

```
        }
        else if(type_of_conn.contains("commercial"))
        {
            if (unit_consumed <= 100)
                tbill=2* unit_consumed;
            else if (unit_consumed >100 && unit_consumed <= 200)
                tbill=4.50* unit_consumed;
            else if (unit_consumed >200 && unit_consumed <= 500)
                tbill=6* unit_consumed;
            else
                tbill=7* unit_consumed;
        }
    }
    void Display()
    {
        System.out.println ("\n\t Customer name = "+cname);
        System.out.println ("\n\t Total units = "+unit_consumed);
        System.out.println ("\n\t Total bill = Rs "+tbill);
    }
}
```

OUTPUT:

```
D:\Java Programs>javac ElectBill.java
D:\Java Programs>java ElectBill
    Enter Consumer number = 102
    Enter Consumer name = Raghav
    Enter the type of connection = domestic
    Enter pre_month reading = 150
    Enter current_month reading = 800
    Customer name = Raghav
    Total units = 650.0
    Total bill = Rs 3900.0

D:\raghu\Java Programs>javac ElectBill.java
D:\raghu\Java Programs>java ElectBill
    Enter Consumer number = 103
    Enter Consumer name = Raj
    Enter the type of connection = commercial
    Enter pre_month reading = 1005
    Enter current_month reading = 1300
    Customer name = Raj
    Total units = 295.0
    Total bill = Rs 1770.0
```

PROGRAM:

```
package currency;
import java.util.*;
import java.text.DecimalFormat;
Class CurrencyConvertor
{
    double rupee,dollar,euro,yen;
    Scanner sc = new Scanner(System.in);
    DecimalFormat f = new DecimalFormat("##.###");
    public convertInrToEuro()
    {
        System.out.println("Enter amount in rupees");
        rupee = sc.nextFloat();
        euro = rupee / 80;
        System.out.println("Euro : "+f.format(euro));
    }
    public convertEuroToInr()
    {
        System.out.println("Enter amount in Euro");
        euro = sc.nextFloat();
        rupee = euro * 80;
        System.out.println("Rupees : "+f.format(rupee));
    }
    public convertInrToDollar()
    {
        System.out.println("Enter amount in rupees");
        rupee = sc.nextFloat();
        dollar = rupee / 66;
        System.out.println("Dollar : "+f.format(dollar));
    }
    public convertDollarToInr()
    {
        System.out.println("Enter amount in Dollar");
        dollar = sc.nextFloat();
        rupee = dollar * 66;
        System.out.println("Rupees : "+f.format(rupee))
    }
    public convertInrToYen()
    {
        System.out.println("Enter amount in rupees");
        rupee = sc.nextFloat();
        yen = rupee / 0.61;
        System.out.println("Yen : "+f.format(yen));
    }
    public convertYenToInr()
    {
```

```
        System.out.println("Enter amount in Yen");
        euro = sc.nextFloat();
        rupee = yen * 0.61;
        System.out.println("Rupees : "+f.format(rupee));
    }
}
```

```
package distance;
import java.util.*;
import java.text.DecimalFormat;
Class DistanceConvertor
{
    double meter,km,miles;
    Scanner sc = new Scanner(System.in);
    DecimalFormat f = new DecimalFormat("##.###");
    public convertMeterToKm()
    {
        System.out.println("Enter the meter");
        meter = sc.nextFloat();
        km =    meter * 0.001;
        System.out.println("Kilometer : "+f.format(km));
    }
    public convertKmToMeter()
    {
        System.out.println("Enter the Kilometer");
        km = sc.nextFloat();
        meter = km / 0.001;
        System.out.println("Meter : "+f.format(meter));
    }
    public convertMilesToKm()
    {
        System.out.println("Enter the miles");
        miles = sc.nextFloat();
        km = miles * 1.6093;
        System.out.println("Kilometer : "+f.format(km));
    }
    public convertKmToMiles()
    {
        System.out.println("Enter the Kilometer");
        km = sc.nextFloat();
        miles = km / 1.6093;
        System.out.println("Miles : "+f.format(miles));
    }
}
```

```
package time;
import java.util.*;
import java.text.DecimalFormat;
Class TimeConvertor
{
    double hour,minute,second;
    Scanner sc = new Scanner(System.in);
    DecimalFormat f = new DecimalFormat("##.###");
    public convertHourToMinute()
    {
        System.out.println("Enter the Hour");
        hour = sc.nextFloat();
        minute = hour * 60;
        System.out.println("Minutes : "+f.format(minute));
    }
    public convertMinuteToHour()
    {
        System.out.println("Enter the Minute");
        minute = sc.nextFloat();
        hour = minute / 60;
        System.out.println("Hours : "+f.format(hour));
    }
    public convertHourToSeconds()
    {
        System.out.println("Enter the Hour");
        hour = sc.nextFloat();
        second = hour * 3600;
        System.out.println("Seconds : "+f.format(second));
    }
    public convertSecondsToHour()
    {
        System.out.println("Enter the Seconds");
        second = sc.nextFloat();
        hour = second / 3600;
        System.out.println("Hours : "+f.format(hour));
    }
}


import currency.*;
import distance.*;
import time.*;
import java.util.Scanner;
public class Convertor
{
    public static void main (String[] args)
```

```
{
    int code,currency_code,distance_code,time_code;
    Scanner sc = new Scanner(System.in);
    System.out.println("Enter the code 1:Currency\n2:Distance\n3:Time");
    code=sc.nextInt();
    if(code==1)
    {
        System.out.println("Enter the Currecy code 1:Euro\n2:Dollar\n3:Yen");
        currency_code=sc.nextInt();
        if(currency_code == 1)
        {
            convertInrToEuro();
            convertEuroToInr()
        }
        else if(currency_code == 2)
        {
            convertInrToDollar();
            convertDollarToInr();
        }
        else if(currency_code == 3)
        {
            convertInrToYen();
            convertYenToInr();
        }
    else
        {
        System.out.println(“Invalid Code”);
        }
    }
    else if(code==2)
    {
        System.out.println("Enter the Distance code 1:Meter\n2:Miles");
        distance_code=sc.nextInt();
        if(distance_code == 1)
        {
            convertMeterToKm();
            convertKmToMeter();
        }
        else if(distance_code == 2)
        {
                convertMilesToKm();
                convertKmToMiles();
        }
        else
        {
            System.out.println(“Invalid Code”);
```

```
            }
        }
        else if(code==3)
        {
            System.out.println("Enter the Time code 1:Minutes\n2:Seconds");
            time_code=sc.nextInt();
            if(time_code == 1)
            {
                convertHourToMinute();
                convertMinuteToHour();
            }
            else if(time_code == 2)
            {
                    convertHourToSeconds();
                    convertSecondsToHour();
            }
            else
            {
                System.out.println(“Invalid Code”);
            }
        }
        else
        {
            System.out.println(“Invalid Code”);
        }
    }
}
```

OUTPUT:

```
Enter the code 1:Currency\n2:Distance\n3:Time
1
Enter the Currecy code 1:Euro\n2:Dollar\n3:Yen
2
Enter amount in rupees
6600
Dollar : 100
Enter amount in Dollar
6
Rupees : 396
```

PROGRAM:

```
import java.io.*;
import java.lang.*;
import java.util.*;
class Employee
{
    String emp_name;
    int emp_id;
    String address;
    String mail_id;
    String mob_number;
    Employee(String emp_name, int emp_id, String address, String mail_id, String
    mob_number)
    {
        this. emp_name = emp_name;
        this.emp_id = emp_id;
        this.address = address;
        this.mob_number =mob_number;
    }
    void display()
    {
        System.out.println("Employee Name: "+ emp_name);
        System.out.println("Employee_ID: "+ emp_id);
        System.out.println("Address: "+ address);
        System.out.println("Mobile Number: "+ mob_number);
    }
}
class Programmer extends Employee
{
    double basicpay,da,hra,pf,fund,netsalary,grosssalary;
    Programmer (String emp_name, int emp_id, String address, String mail_id, String
    mob_number,double bp)
    {
        super(emp_name, emp_id , address, mail_id, mob_number);
        basicpay = bp;
    }
    public void getPaySlip()
    {
        da=basicpay * 97/100;
        hra=basicpay* 10/100;
        pf=basicpay* 12/100;
        fund= basicpay* 0.1/100;
        grosssalary=basicpay+da+hra+pf+fund;
        netsalary = grosssalary-pf-fund;
    }
    void disp()
```

```
    {
        System.out.println("Name of The Employee :"+ emp_name +"***** payslip ****");
        display();
        System.out.println(" Grosssalary ="+ grosssalary);
        System.out.println(" Netsalary ="+ netsalary);
    }
}
class AssistantProfessor extends Employee
{
    double basicpay,da,hra,pf,fund,netsalary,grosssalary;
    AssistantProfessor (String emp_name, int emp_id, String address, String mail_id, String
    mob_number,double bp)
    {
        super(emp_name, emp_id , address, mail_id, mob_number);
        basicpay = bp;
    }
    public void getPaySlip()
    {
        da=basicpay * 97/100;
        hra=basicpay* 10/100;
        pf=basicpay* 12/100;
        fund= basicpay* 0.1/100;
        grosssalary=basicpay+da+hra+pf+fund;
        netsalary = grosssalary-pf-fund;
    }
    void disp()
    {
        System.out.println("Name of The Employee :"+ emp_name +"***** payslip ****");
        display();
        System.out.println(" Grosssalary ="+ grosssalary);
        System.out.println(" Netsalary ="+ netsalary);
    }
}
class AssociateProfessor extends Employee
{
    double basicpay,da,hra,pf,fund,netsalary,grosssalary;
    AssociateProfessor (String emp_name, int emp_id, String address, String mail_id, String
    mob_number,double bp)
    {
        super(emp_name, emp_id , address, mail_id, mob_number);
        basicpay = bp;
    }
    public void getPaySlip()
    {
        da=basicpay * 97/100;
        hra=basicpay* 10/100;
```

```
        pf=basicpay* 12/100;
        fund= basicpay* 0.1/100;
        grosssalary=basicpay+da+hra+pf+fund;
        netsalary = grosssalary-pf-fund;
    }
    void disp()
    {
        System.out.println("Name of The Employee :"+ emp_name +"***** payslip ****");
        display();
        System.out.println(" Grosssalary ="+ grosssalary);
        System.out.println(" Netsalary ="+ netsalary);
    }
}
class Professor extends Employee
{
    double basicpay,da,hra,pf,fund,netsalary,grosssalary;
    Professor (String emp_name, int emp_id, String address, String mail_id, String
    mob_number,double bp)
    {
        super(emp_name, emp_id , address, mail_id, mob_number);
        basicpay = bp;
    }
    public void getPaySlip()
    {
        da=basicpay * 97/100;
        hra=basicpay* 10/100;
        pf=basicpay* 12/100;
        fund= basicpay* 0.1/100;
        grosssalary=basicpay+da+hra+pf+fund;
        netsalary = grosssalary-pf-fund;
    }
    void disp()
    {
        System.out.println("Name of The Employee :"+ emp_name +"***** payslip ****");
        display();
        System.out.println("Grosssalary ="+ grosssalary);
        System.out.println("Netsalary ="+ netsalary);
    }
}
public class EmployeePayslip
{
    public static void main(String arg[])throws IOException
    {
        String name,add,mail,mob;
        int id,desg;
        double bp;
```

```
        DataInputStream in = new DataInputStream(System.in);
        System.out.println("Enter Name of Employee :");
        name=in.readLine();
        System.out.println("Enter ID of Employee :");
        id= Integer.valueOf(in.readLine());
        System.out.println("Enter Address of Employee :");
        add=in.readLine();
        System.out.println("Enter Mail ID of Employee :");
        mail=in.readLine();
        System.out.println("Enter Mobile Number of Employee :");
        mob=in.readLine();
        System.out.println("Enter the Basicpay :");
        bp=Double.valueOf(in.readLine());
        System.out.println("Enter the Designation :");
        System.out.println("1. Programmer\n2. AssistantProfessor\n 3.
        AssociateProfessor\n4.Professor \n5. Exit");
        desg= Integer.valueOf(in.readLine());
        switch (desg)
        {
        case 1: Programmer p=new Programmer (name,id,add,mail,mob,bp);
                p. getPaySlip ();
                p.disp();
                break;
        case 2: AssistantProfessor ap=new AssistantProfessor(name,id,add,mail,mob,bp);
                ap. getPaySlip ();
                ap.disp(); break;
        case 3: AssociateProfessor assp=new AssociateProfessor(name,id,add,mail,mob,bp);
                assp. getPaySlip ();
                assp.disp();
                break;
        case 4: Professor pf=new Professor (name,id,add,mail,mob,bp);
                pf. getPaySlip ();
                pf.disp();
                break;
        case 5: System.exit(0);
        default: System.out.println("Invalid designation");
        }
    }
}
```

OUTPUT:

D:\ Java Programs>java EmployeePayslip
Enter Name of Employee :
Raj
Enter ID of Employee :
327
Enter Address of Employee :
Villupuram
Enter Mail ID of Employee :
raj@gmail.com
Enter Mobile Number of Employee :
9994191599
Enter the Basicpay :
15000
Enter the Designation :

1. Programmer
2. AssistantProfessor
3. AssociateProfessor
4. Professor
5. Exit

2
Name of The Employee :Raghu***** payslip ****
Employee Name: Raghu
Employee_ID: 327
Address: Villupuram
Mobile Number: 9994191599
Grosssalary =32865.0
Netsalary =31050.0

PROGRAM:

```
import java.util.*;
abstract class shape
{
    public int x,y;
    public abstract void printArea();
}
class Rectangle1 extends shape
{
    public void printArea() {
        float area;
        area= x * y;
        System.out.println("Area of Rectangle is " +area);
    }
}
class Triangle extends shape
{
    public void printArea()
    {
        float area;
        area= (x * y) / 2;
        System.out.println("Area of Triangle is " + area);
    }
}
class Circle extends shape
{
    public void printArea()
    {
        float area;
        area=(22 * x * x) / 7;
        System.out.println("Area of Circle is " + area);
    }
}
public class Shapes
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter values : ");
        int x1=sc.nextInt();
        int y1=sc.nextInt();
        Rectangle1 r = new Rectangle1();
        r.x = x1; r.y = y1; r.printArea();
        Triangle t = new Triangle();
        t.x = x1; t.y = y1; t.printArea();
        Circle c = new Circle();
```

```
            c.x = x1;
            c.printArea();
        }
}
```

OUTPUT:

```
D:\JavaPrograms>java Shapes
Enter values :
7
8
Area of Rectangle is 56.0
Area of Triangle is 28.0
Area of Circle is 154.0
```

PROGRAM:

```
import java.io.*;
interface Mystack
{
    public void pop();
    public void push();
    public void display();
}
class Stack_array implements Mystack
{
    final static int n=5;
    int stack[]=new int[n];
    int top=-1;
    public void push()
    {
        try
        {
            BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
            if(top==(n-1))
            {
                System.out.println(" Stack Overflow");
                return;
            }
            else
            {
                System.out.println("Enter the element");
                int ele=Integer.parseInt(br.readLine());
                stack[++top]=ele;
            }
        }
        catch(IOException e)
        {
            System.out.println("e");
        }
    }
    public void pop()
    {
        if(top<0)
        {
            System.out.println("Stack underflow");
            return;
        }
        else
        {
            int popper=stack[top];
            top--;
```

```
            System.out.println("Popped element:" +popper);
        }
    }
    public void display()
    {
        if(top<0)
        {
            System.out.println("Stack is empty"); return;
        }
        else
        {
            String str=" ";
            for(int i=0; i<=top; i++) {
                str=str+" "+stack[i]+" <--";}
            System.out.println("Elements are:"+str);
        }
    }
}
class StackADT
{
    public static void main(String arg[])throws IOException
    {
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
        System.out.println("Implementation of Stack using Array");
        Stack_array stk=new Stack_array(); int ch=0;
        do
        {
            System.out.println("1.Push 2.Pop 3.Display 4.Exit);
            System.out.println("Enter your choice:");
            ch=Integer.parseInt(br.readLine());
            switch(ch)
            {
            case 1:
                stk.push();
                break;
            case 2:
                stk.pop();
                break;
            case 3:
                stk.display();
                break;
            case 4:
                System.exit(0);
            }
        }
        while(ch<5);
```

```
    }
}
```

Output

```
Implementation of Stack using Array
1.Push 2.Pop 3.Display 4.Exit
Enter your choice:
1
Enter the element
10
1.Push 2.Pop 3.Display 4.Exit
Enter your choice:
1
Enter the element
15
1.Push 2.Pop 3.Display 4.Exit
Enter your choice:
1
Enter the element
25
1.Push 2.Pop 3.Display 4.Exit
Enter your choice:
3
Elements are: 10 <-- 15 <-- 25 <--
1.Push 2.Pop 3.Display 4.Exit
2
Popped element: 25
```

PROGRAM:

```
public class WrapperClassDemo
{
	public static void main(String[] args)
	{
		int primitiveInt = 100;
		float primitiveFloat = 25.75f;
		char primitiveChar = 'A';
		boolean primitiveBoolean = true;
		Integer wrappedInt = primitiveInt;
		Float wrappedFloat = primitiveFloat;
		Character wrappedChar = primitiveChar;
		Boolean wrappedBoolean = primitiveBoolean;
		System.out.println("Autoboxed Integer: " + wrappedInt);
		System.out.println("Autoboxed Float: " + wrappedFloat);
		System.out.println("Autoboxed Character: " + wrappedChar);
		System.out.println("Autoboxed Boolean: " + wrappedBoolean);
		int unboxedInt = wrappedInt;
		float unboxedFloat = wrappedFloat;
		char unboxedChar = wrappedChar;
		boolean unboxedBoolean = wrappedBoolean;
		System.out.println("Unboxed int: " + unboxedInt);
		System.out.println("Unboxed float: " + unboxedFloat);
		System.out.println("Unboxed char: " + unboxedChar);
		System.out.println("Unboxed boolean: " + unboxedBoolean);
		String intString = "300";
		Integer parsedInt = Integer.parseInt(intString);
		String floatString = "75.25";
		Float parsedFloat = Float.parseFloat(floatString);
		String charString = "C";
		Character parsedChar = charString.charAt(0);
		String booleanString = "true";
		Boolean parsedBoolean = Boolean.parseBoolean(booleanString);
		System.out.println("Parsed and autoboxed Integer: " + parsedInt);
		System.out.println("Parsed and autoboxed Float: " + parsedFloat);
		System.out.println("Parsed and autoboxed Character: " + parsedChar);
		System.out.println("Parsed and autoboxed Boolean: " + parsedBoolean);
	}
}
```

OUTPUT:

```
Autoboxed Integer: 100
Autoboxed Float: 25.75
Autoboxed Character: A
Autoboxed Boolean: true
Unboxed int: 100
Unboxed float: 25.75
Unboxed char: A
Unboxed boolean: true
Parsed and autoboxed Integer: 300
Parsed and autoboxed Float: 75.25
Parsed and autoboxed Character: C
Parsed and autoboxed Boolean: tr
```

PROGRAM:

```
import java.util.*;
class even implements Runnable
{
    public int x;
    public even(int x)
    {
        this.x = x;
    }
    public void run()
    {
        System.out.println("New Thread "+ x +" is EVEN and Square of " + x + " is: " + x * x);
    }
}
class odd implements Runnable
{
    public int x;
    public odd(int x)
    {
        this.x = x;
    }
    public void run()
    {
        System.out.println("New Thread "+ x +" is ODD and Cube of " + x + " is: " + x * x * x);
    }
}
class A extends Thread
{
    public void run()
    {
        int num = 0;
        Random r = new Random();
        try
        {
            for (int i = 0; i < 5; i++)
            {
                num = r.nextInt(100);
                System.out.println("Main Thread and Generated Number is " + num);
                if (num % 2 == 0)
                {
                    Thread t1 = new Thread(new even(num));
                    t1.start();
                }
                else
                {
                    Thread t2 = new Thread(new odd(num));
```

```
                    t2.start();
                }
                Thread.sleep(1000);
                System.out.println("----------------------------------------");
            }
        }
        catch (Exception ex)
        {
            System.out.println(ex.getMessage());
        }
    }
}
public class ThreadProgram
{
    public static void main(String[] args)
    {
        A a = new A();
        a.start();
    }
}
```

OUTPUT:

```
D:\JP>java ThreadProgram
Main Thread and Generated Number is 10
New Thread 10 is EVEN and Square of 10 is: 100
----------------------------------------

Main Thread and Generated Number is 14
New Thread 14 is EVEN and Square of 14 is: 196
----------------------------------------

Main Thread and Generated Number is 83
New Thread 83 is ODD and Cube of 83 is: 571787
----------------------------------------

Main Thread and Generated Number is 1
New Thread 1 is ODD and Cube of 1 is: 1
----------------------------------------

Main Thread and Generated Number is 20
New Thread 20 is EVEN and Square of 20 is: 400
```

PROGRAM:

```
class SharedResource {
    private int data;
    private boolean available = false;
    public synchronized void produce(int value) {
        while (available) {
            try {
                wait();
            }
            catch (InterruptedException e) {
                System.out.println("Thread interrupted");
            }
        }
        data = value;
        System.out.println("Produced: " + data);
        available = true;
        notify();
    }
    public synchronized void consume() {
        while (!available) {
            try {
                wait();
            }
            catch (InterruptedException e) {
                System.out.println("Thread interrupted");
            }
        }
        System.out.println("Consumed: " + data);
        available = false;
        notify();
    }
}
class Producer implements Runnable {
    private SharedResource resource;
    public Producer(SharedResource resource) {
        this.resource = resource;
    }
    public void run() {
        for (int i = 1; i <= 5; i++) {
            resource.produce(i);
            try {
                Thread.sleep(100);
            }
            catch (InterruptedException e) {
                System.out.println("Thread interrupted");
            }
```

```
        }
    }
}
class Consumer implements Runnable {
    private SharedResource resource;
    public Consumer(SharedResource resource) {
        this.resource = resource;
    }
    public void run() {
        for (int i = 1; i <= 5; i++) {
            resource.consume();
            try {
                Thread.sleep(100);
            }
            catch (InterruptedException e) {
                System.out.println("Thread interrupted");
            }
        }
    }
}
public class InterThreadCommunication {
    public static void main(String[] args) {
        SharedResource resource = new SharedResource();
        Thread producerThread = new Thread(new Producer(resource));
        Thread consumerThread = new Thread(new Consumer(resource));
        producerThread.start();
        consumerThread.start();
    }
}
```

OUTPUT:

```
Produced: 1
        Consumed: 1
        Produced: 2
        Consumed: 2
        Produced: 3
        Consumed: 3
        Produced: 4
        Consumed: 4
        Produced: 5
        Consumed: 5
```

PROGRAM:

```
import java.util.*;
public class ArrayListExample {
    public static void main(String args[]) {
        /*Creation of ArrayList: I'm going to add String elements so I made it of string type */
        ArrayList<String> obj1 = new ArrayList<String>();
        /*This is how elements should be added to the array list*/
        obj1.add("Ajeet");
        obj1.add("Harry");
        obj1.add("Chaitanya");
        obj1.add("Steve");
        obj1.add("Anuj");
        /* Displaying array list elements */
        System.out.println("\nCurrently the array list obj1 has following elements:"+obj1);
        /*Append element at the end*/
        obj1.add("Babu");
        obj1.add("Kamal");
        /*Append collection element to ArrayList*/
        ArrayList<String> obj2 = new ArrayList<String>();
        obj2.add("Alice");
        obj2.add("Bob");
        obj2.add("Raj");
        //Adding ArrayList obj2 into ArrayList obj1
        obj1.addAll(obj2);
        System.out.println("\nArrayList obj1 after add All:"+obj1);
        /*Add element at the given index*/
        obj1.add(0, "Rahul");
        obj1.add(1, "Justin");
        System.out.println("\nArrayList obj1 after add element at the given index:"+obj1);
        /*Search an element*/
        System.out.println("\nEnter the Search element:");
        Scanner input=new Scanner(System.in);
        String search=input.nextLine();
        System.out.println("\nArrayList obj1 contains the string "+search+"
        :"+obj1.contains(search));
        /*Remove elements from array list like this*/
        obj1.remove("Chaitanya");
        obj1.remove("Harry");
        System.out.println("\nCurrent array list of obj1 after removing element is:"+obj1);
        /*Remove element from the given index*/
        obj1.remove(1);
        System.out.println("\nCurrent array list of obj1 after removing element through index
        is:"+obj1);
        System.out.println("\nEnter the letter to display all the string start with given letter:");
        search=input.nextLine();
        ArrayList<String> obj3 = new ArrayList<String>();
```

```
        for(int i=0;i<obj1.size();i++)
        {
            if(obj1.get(i).startsWith(search.toUpperCase()))
            {
                obj3.add(obj1.get(i));
            }
        }
        if(obj3.size()>0)
        {
            System.out.println("\nArrayList obj1 contains all the string start with given
            "+search+":"+obj3);
        }
        else
        {
            System.out.println("\nNo Name start with "+search+" letter in Arraylist obj1");
        }
    }
}
```

OUTPUT:

```
D:\raghu\Java Programs>java ArrayListExample
Currently the array list obj1 has following elements:[Ajeet, Harry, Chaitanya, Steve, Anuj]
ArrayList obj1 after add All:[Ajeet, Harry, Chaitanya, Steve, Anuj, Babu, Kamal, Alice, Bob,
Raj]
ArrayList obj1 after add element at the given index:[Rahul, Justin, Ajeet, Harry, Chaitanya,
Steve, Anuj, Babu, Kamal, Alice, Bob, Raj]
Enter the Search element:
Babu
ArrayList obj1 contains the string Babu :true
Current array list of obj1 after removing element is:[Rahul, Justin, Ajeet, Steve, Anuj, Babu,
 Kamal, Alice, Bob, Raj]
Current array list of obj1 after removing element through index is:[Rahul, Ajeet, Steve, Anuj,
Babu, Kamal, Alice, Bob, Raj]
Enter the letter to display all the string start with given letter:
R
ArrayList obj1 contains all the string start with given
```

PROGRAM:

```
import java.util.Scanner;
import java.io.File;
class FileDemo
{
    public static void main(String[] args)
    {
        Scanner input=new Scanner(System.in);
        String s=input.nextLine();
        File f1=new File(s);
        System.out.println("File Name:"+f1.getName());
        System.out.println("Path:"+f1.getPath());
        System.out.println("Abs Path:"+f1.getAbsolutePath());
        System.out.println("Parent:"+f1.getParent());
        System.out.println("This file is:"+(f1.exists()?"Exists":"Does not exists"));
        System.out.println("Is file:"+f1.isFile());
        System.out.println("Is Directory:"+f1.isDirectory());
        System.out.println("Is Readable:"+f1.canRead());
        System.out.println("IS Writable:"+f1.canWrite());
        System.out.println("Is Absolute:"+f1.isAbsolute());
        System.out.println("File Last Modified:"+f1.lastModified());
        System.out.println("File Size:"+f1.length()+"bytes");
        System.out.println("Is Hidden:"+f1.isHidden());
    }
}
```

OUTPUT:

```
D:/JavaPrograms> java FileDemo
Fibonacci.java
File Name:Fibonacci.java
Path: Fibonacci.java
Abs Path: c:\sameer\Fibonacci.java
Parent: Null
This file is:Exists
Is file:true
Is Directory:false
Is Readable:true
Is Writable:true
Is Absolute:false
File Last Modified:1206324301937
File Size: 406 bytes
Is Hidden: false
```

MySQL COMMANDS:

```
CREATE DATABASE studentdb;
USE studentdb;
CREATE TABLE students (
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100) NOT NULL,
age INT NOT NULL,
course VARCHAR(100) NOT NULL
);
```

PROGRAM:

```
import javafx.application.Application;
import javafx.geometry.Insets;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.GridPane;
import javafx.scene.paint.Color;
import javafx.scene.text.Font;
import javafx.stage.Stage;
import java.sql.*;
public class StudentManagementApp extends Application {
        private Connection connection;
        private TextField idField = new TextField();
        private TextField nameField = new TextField();
        private TextField ageField = new TextField();
        private TextField courseField = new TextField();
        private TextArea displayArea = new TextArea();
        public static void main(String[] args) {
            launch(args);
        }
        public void start(Stage primaryStage) {
                connectToDatabase();
                // Set up the layout
                GridPane gridPane = new GridPane();
                gridPane.setPadding(new Insets(10));
                gridPane.setHgap(10);
                gridPane.setVgap(10);
                 // Set up font
                 Font font = new Font("Arial", 14);
                 idField.setPromptText("ID (for Update/Delete)");
                 idField.setFont(font);
                 nameField.setPromptText("Name");
                 nameField.setFont(font);
                 ageField.setPromptText("Age");
                 ageField.setFont(font);
                 courseField.setPromptText("Course");
```

```
        courseField.setFont(font);
        Button createButton = new Button("Create");
        createButton.setFont(font);
        createButton.setStyle("-fx-background-color: #4CAF50; -fx-text-fill:
white;");
        createButton.setOnAction(e -> createStudent());
        Button readButton = new Button("Display");
        readButton.setFont(font);
        readButton.setStyle("-fx-background-color: #2196F3; -fx-text-fill:
white;");
        readButton.setOnAction(e -> readStudents());
        Button updateButton = new Button("Update");
        updateButton.setFont(font);
        updateButton.setStyle("-fx-background-color: #FF9800; -fx-text-fill:
white;");
        updateButton.setOnAction(e -> updateStudent());
        Button deleteButton = new Button("Delete");
        deleteButton.setFont(font);
        deleteButton.setStyle("-fx-background-color: #F44336; -fx-text-fill:
white;");
        deleteButton.setOnAction(e -> deleteStudent());
        displayArea.setFont(font);
        displayArea.setEditable(false);
        displayArea.setWrapText(true);
        // Arrange elements in the grid
        gridPane.add(new Label("ID:"), 0, 0);
        gridPane.add(idField, 1, 0);
        gridPane.add(new Label("Name:"), 0, 1);
        gridPane.add(nameField, 1, 1);
        gridPane.add(new Label("Age:"), 0, 2);
        gridPane.add(ageField, 1, 2);
        gridPane.add(new Label("Course:"), 0, 3);
        gridPane.add(courseField, 1, 3);
        // Arrange buttons in two rows with gaps
        gridPane.add(createButton, 0, 4);
        gridPane.add(readButton, 1, 4);
        gridPane.add(updateButton, 0, 5);
        gridPane.add(deleteButton, 1, 5);
        GridPane.setMargin(createButton, new Insets(5, 5, 5, 5));
        GridPane.setMargin(readButton, new Insets(5, 5, 5, 5));
        GridPane.setMargin(updateButton, new Insets(5, 5, 5, 5));
        GridPane.setMargin(deleteButton, new Insets(5, 5, 5, 5));
         // Add display area
        gridPane.add(displayArea, 0, 6, 2, 1);
         // Set up the scene
        Scene scene = new Scene(gridPane, 400, 500);
```

```
                primaryStage.setTitle("Student Management");
                primaryStage.setScene(scene);
                primaryStage.show();
        }
        public void connectToDatabase() {
                try {
                    // Load the MySQL JDBC driver
                    Class.forName("com.mysql.cj.jdbc.Driver");
                    // Establish a connection to the database
                    connection =
        DriverManager.getConnection("jdbc:mysql://localhost:3306/studentdb",
        "root", "1234");
                    System.out.println("Database connection successful");
                }
                catch (ClassNotFoundException e) {
                    System.err.println("MySQL JDBC Driver not found");
                    e.printStackTrace();
                } catch (SQLException e) {
                    System.err.println("Database connection failed");
                    e.printStackTrace();
            }
        }
        private void createStudent() {
                String name = nameField.getText();
                int age = Integer.parseInt(ageField.getText());
                String course = courseField.getText();
                String sql = "INSERT INTO students (name, age, course) VALUES (?, ?, ?)";
                try (PreparedStatement pstmt = connection.prepareStatement(sql)) {
                    pstmt.setString(1, name);
                    pstmt.setInt(2, age);
                    pstmt.setString(3, course);
                    pstmt.executeUpdate();
                    displayArea.setText("Student created successfully.");
                }
                catch (SQLException e) {
                        e.printStackTrace();
                }
        }
        private void readStudents() {
                String sql = "SELECT * FROM students";
                try (Statement stmt = connection.createStatement();
                ResultSet rs =      stmt.executeQuery(sql)) {
                    StringBuilder sb = new StringBuilder();
                    while (rs.next()) {
                            sb.append("ID: ").append(rs.getInt("id"))
                                    .append(", Name: ").append(rs.getString("name"))
```

```
                                .append(", Age: ").append(rs.getInt("age"))
                                .append(", Course: ").append(rs.getString("course"))
                                .append("\n");
                    }
                    displayArea.setText(sb.toString());
                } catch (SQLException e) {
                    e.printStackTrace();
                }
        }
        private void updateStudent() {
                int id = Integer.parseInt(idField.getText());
                String name = nameField.getText();
                int age = Integer.parseInt(ageField.getText());
                String course = courseField.getText();
                String sql = "UPDATE students SET name = ?, age = ?, course = ? WHERE
    id = ?";
                try (PreparedStatement pstmt = connection.prepareStatement(sql)) {
                        pstmt.setString(1, name);
                        pstmt.setInt(2, age);
                        pstmt.setString(3, course);
                        pstmt.setInt(4, id);
                        pstmt.executeUpdate();
                        displayArea.setText("Student updated successfully.");
                } catch (SQLException e) {
                        e.printStackTrace();
                }
        }
        private void deleteStudent() {
                int id = Integer.parseInt(idField.getText());
                String sql = "DELETE FROM students WHERE id = ?";
                try (PreparedStatement pstmt = connection.prepareStatement(sql)) {
                        pstmt.setInt(1, id);
                        pstmt.executeUpdate();
                        displayArea.setText("Student deleted successfully.");
                } catch (SQLException e) {
                        e.printStackTrace();
                }
        }
        @Override
        public void stop() throws Exception {
                if (connection != null && !connection.isClosed()) {
                    connection.close();
                }
                super.stop();
        }
}
```

OUTPUT:

Compilation Command:

G:\IFET\prgs>set CLASSPATH="C:\Program Files\Java\mysql-connector-j-8.4.0\mysql-connector-java-8.4.0.jar;C:\Program Files\Java\openjfx-22.0.1_windows-x64_bin-sdk\javafx-sdk-22.0.1\lib*"

G:\IFET\prgs>javac --module-path "C:\Program Files\Java\openjfx-22.0.1_windows-x64_bin-sdk\javafx-sdk-22.0.1\lib" --add-modules javafx.controls,javafx.fxml StudentManagementApp.java

Run Command:

G:\IFET\prgs>java --module-path "C:\Program Files\Java\openjfx-22.0.1_windows-x64_bin-sdk\javafx-sdk-22.0.1\lib;C:\Program Files\Java\mysql-connector-j-8.4.0\mysql-connector-j-8.4.0.jar" --add-modules javafx.controls,javafx.fxml -cp ".;C:\Program Files\Java\mysql-connector-j-8.4.0\mysql-connector-j-8.4.0.jar" StudentManagementA pp

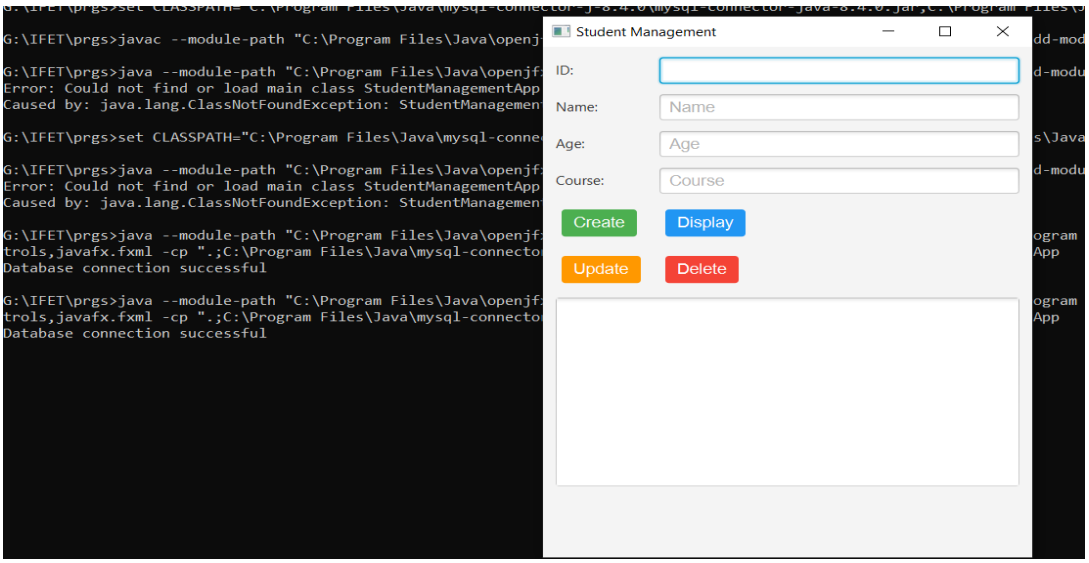


Fig. 1 Student Management Interface

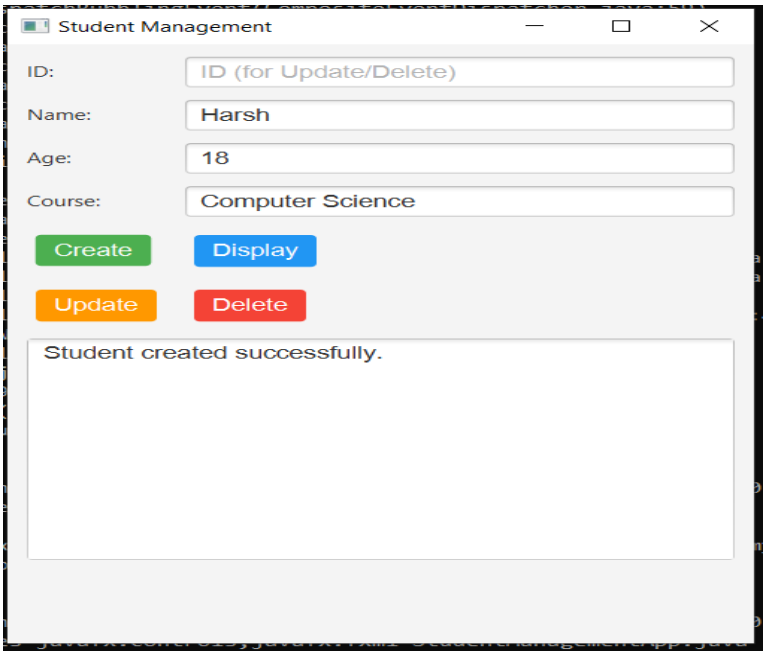


Fig. 2 Data Insertion

```
mysql> use studentdb;
Database changed
mysql> select * from students;
+----+-------+-----+------------------+
| id | name  | age | course           |
+----+-------+-----+------------------+
|  2 | Harsh |  18 | Computer Science |
|  3 | Adhi  |  19 | AIDS             |
|  4 | Nisha |  20 | AIML             |
+----+-------+-----+------------------+
3 rows in set (0.00 sec)

mysql>
```

Fig. 3: Student data in database mysql

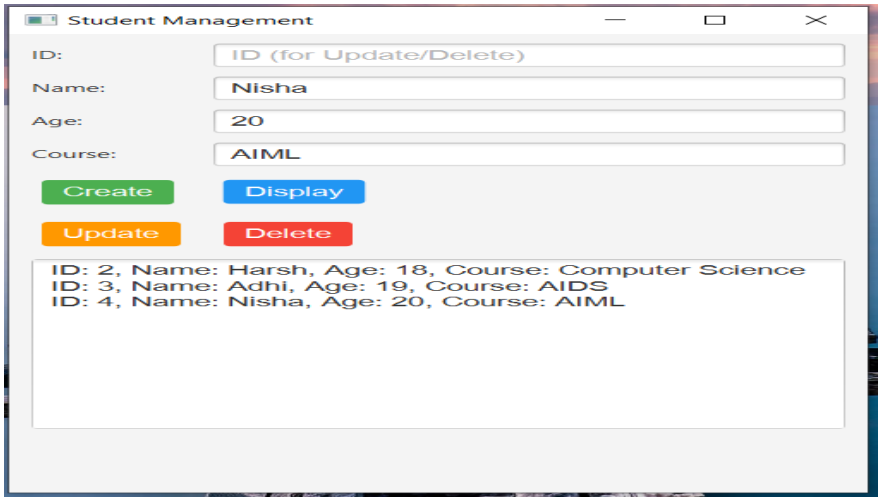


Fig. 4: Display All the data from Database

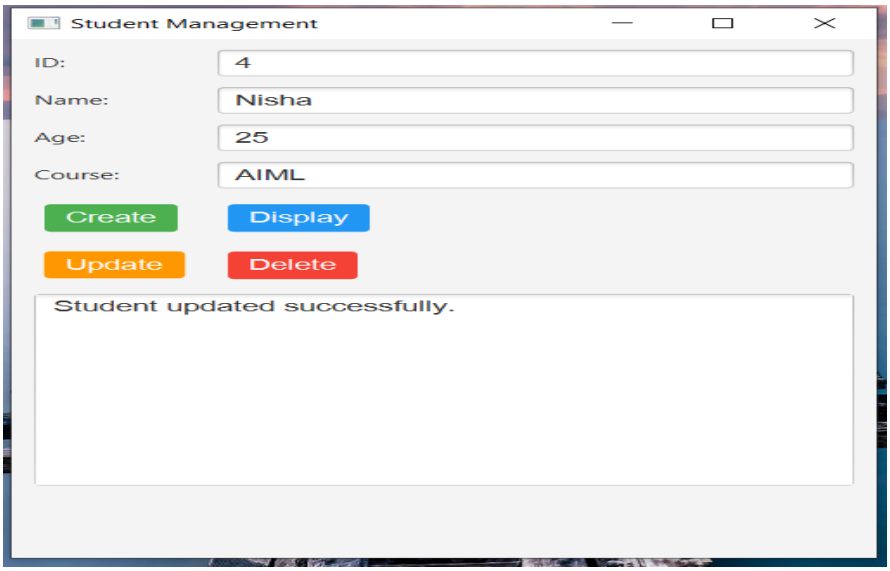


Fig. 5: Updating Student information



Fig. 6: Displaying data after updation

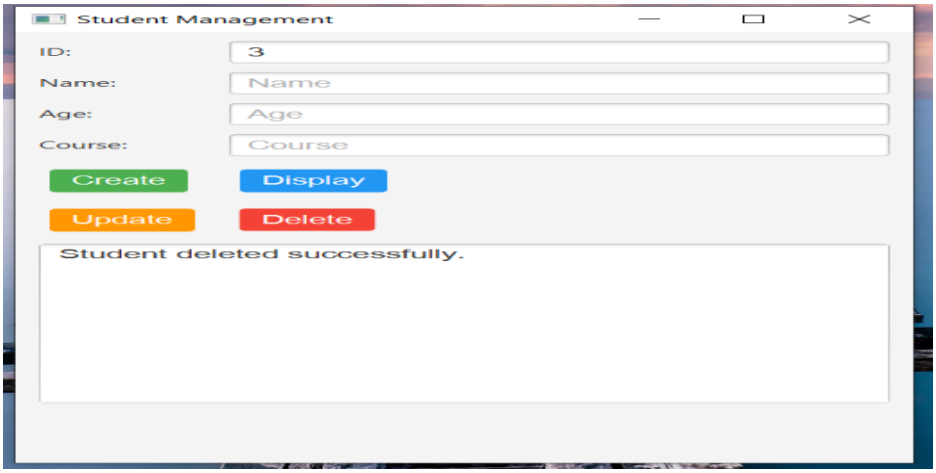


Fig. 7: Deletion

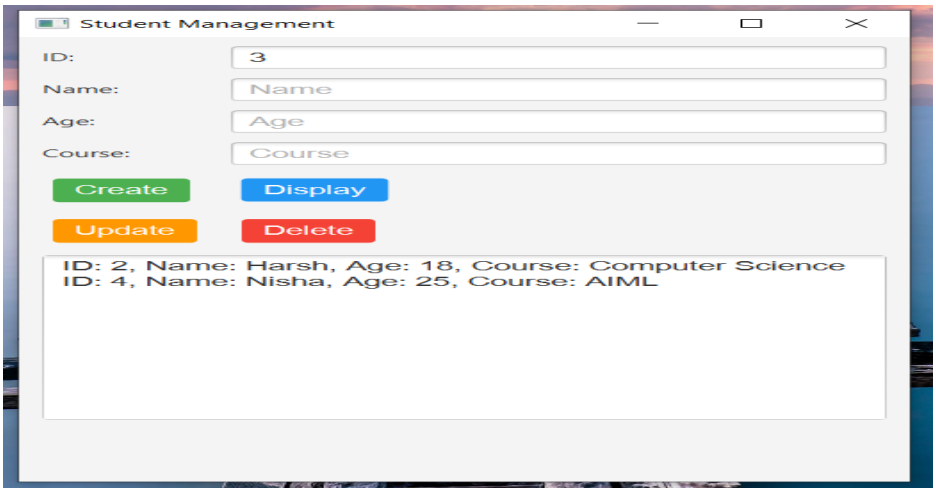


Fig. 8: Displaying information after deletion